

Partial fraction integration

Key Points

Splitting a rational integrand into partial fractions make integration easier.

We can now use the squared denominators (by inverse trig), see the formula sheet

For partial fractions use all factors fractions with numerators of lesser order

Basic Skills

Q1. Write the following as partial fractions

a) $\frac{1}{(x+1)(x+2)}$ b) $\frac{4x^2}{(x+2)(x^2+16)}$

Q2. Using an appropriate integration formula find

a) $\int \frac{1}{x^2+4} dx$ b) $\int \frac{1}{\sqrt{81-x^2}} dx$

Q3. a) Factorise $x^3 - 5x^2 + 6x$

b) Hence write $\frac{6}{x^3-5x^2+6x}$ in partial fractions

c) Hence find $\int \frac{6}{x^3-5x^2+6x} dx$

Q4. Using the formula sheet, or otherwise, find $\int \frac{1}{25-x^2} dx$ in logarithm form

Competency Skills

Q5. Evaluate the following integrals

a) $\int \frac{8}{(x^2+1)(x+1)} dx$ b) $\int \frac{x-6}{(x-1)(x^2+9)} dx$

Q6. Evaluate the integral $\int_3^4 \frac{8x+10}{(x^2+1)(x+4)} dx$

Q7. Find the exact area under the graph $y = \frac{1}{(x^2+5)(x-1)}$ between $x = 2$ and $x = 3$

Q8. a) Using the formula sheet or otherwise state, in logarithm form, $\int \frac{1}{x^2-4} dx$

b) Given that $x + 1$ is a factor of $x^3 + x^2 - 4x - 4$ factorise the expression

c) Hence find the function with gradient $f'(x) = \frac{3}{x^3+x^2-4x-4}$ going through the origin

Advanced Skills

Q9. Given that $y = \frac{4kx}{(x-4k)(x^2+4k^2)}$ for some constant k

a) Express y in partial fractions

b) Hence find $\int y dx$ in terms of k

Q10. a) Express $f(x) = \frac{2}{(1-x)(1+x)(1+x^2)}$ in partial fractions

b) Hence find $\int_0^{\frac{1}{\sqrt{3}}} \frac{8}{1-x^4} dx$

Q11. Find the area under the graph $y = \frac{1}{(x+1)(x^2+1)}$ bound by the positive, positive quadrant

Extension

Q12. Can partial fraction methods be used to evaluate $\int \frac{1}{\sqrt{16-x^2}(x^2+1)} dx$